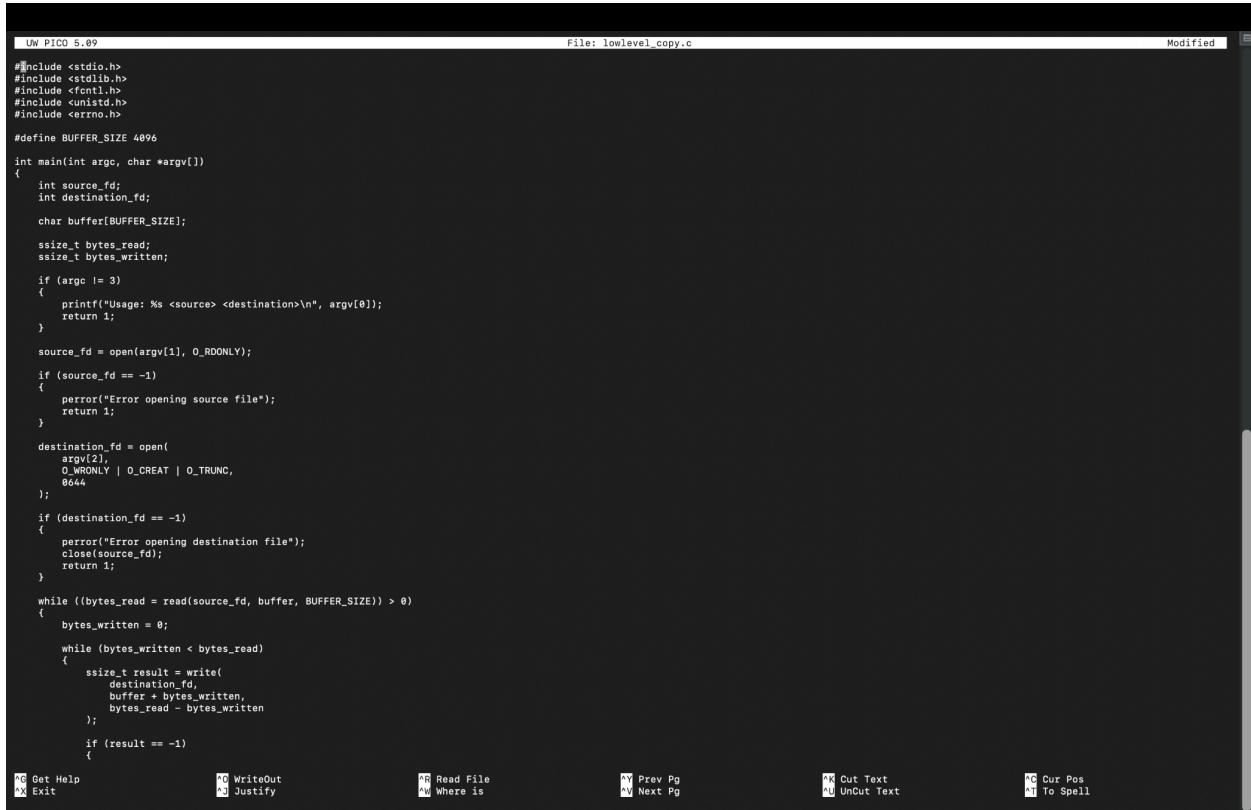


WEEK9-PRACTICAL

Task1:

Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite).



```
UW PICO 5.09 File: lowlevel_copy.c Modified
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <errno.h>

#define BUFFER_SIZE 4096

int main(int argc, char *argv[])
{
    int source_fd;
    int destination_fd;

    char buffer[BUFFER_SIZE];

    ssize_t bytes_read;
    ssize_t bytes_written;

    if (argc != 3)
    {
        printf("Usage: %s <source> <destination>\n", argv[0]);
        return 1;
    }

    source_fd = open(argv[1], O_RDONLY);

    if (source_fd == -1)
    {
        perror("Error opening source file");
        return 1;
    }

    destination_fd = open(
        argv[2],
        O_WRONLY | O_CREAT | O_TRUNC,
        0644
    );

    if (destination_fd == -1)
    {
        perror("Error opening destination file");
        close(source_fd);
        return 1;
    }

    while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0)
    {
        bytes_written = 0;

        while (bytes_written < bytes_read)
        {
            ssize_t result = write(
                destination_fd,
                buffer + bytes_written,
                bytes_read - bytes_written
            );

            if (result == -1)
            {
                perror("Error writing to destination file");
                close(source_fd);
                close(destination_fd);
                return 1;
            }

            bytes_written += result;
        }
    }

    close(source_fd);
    close(destination_fd);

    return 0;
}
```

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```
UW PIC0 5.09 File: lowlevel_copy.c Modified
}
destination_fd = open(
    argv[2],
    O_WRONLY | O_CREAT | O_TRUNC,
    0644
);
if (destination_fd == -1)
{
    perror("Error opening destination file");
    close(source_fd);
    return 1;
}
while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0)
{
    bytes_written = 0;
    while (bytes_written < bytes_read)
    {
        ssize_t result = write(
            destination_fd,
            buffer + bytes_written,
            bytes_read - bytes_written
        );
        if (result == -1)
        {
            perror("Error writing file");
            close(source_fd);
            close(destination_fd);
            return 1;
        }
        bytes_written += result;
    }
}
if (bytes_read == -1)
{
    perror("Error reading file");
}
close(source_fd);
close(destination_fd);
printf("File copied successfully\n");
return 0;
}

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Exit Justify Where is Next Pg UnCut Text To Spell
```

```
Last login: Tue Sep 22 11:21:14 on ttys001
(leva1)8: command not found: >
vishalkarnati@VISHALS-MacBook-Air ~ % cd ~/Desktop/os
vishalkarnati@VISHALS-MacBook-Air os % mkdir week9pract
vishalkarnati@VISHALS-MacBook-Air os % cd week9pract
vishalkarnati@VISHALS-MacBook-Air week9pract % nano lowlevel_copy.c
vishalkarnati@VISHALS-MacBook-Air week9pract % gcc lowlevel_copy.c -o lowlevel_copy
vishalkarnati@VISHALS-MacBook-Air week9pract % nano source.txt
vishalkarnati@VISHALS-MacBook-Air week9pract % cat source.txt
Hello
This is a file copy test.
Operating Systems
Low-level file I/O
vishalkarnati@VISHALS-MacBook-Air week9pract % ./lowlevel_copy source.txt copy.txt
File copied successfully
vishalkarnati@VISHALS-MacBook-Air week9pract % cat copy.txt
Hello
This is a file copy test.
Operating Systems
Low-level file I/O
vishalkarnati@VISHALS-MacBook-Air week9pract %
```

//Standard Library Copy

```
UW PICO 5.09 File: library_copy.c Modified
#include <stdio.h>
#include <stdlib.h>

#define BUFFER_SIZE 4096

int main(int argc, char *argv[])
{
    FILE *source;
    FILE *destination;

    char buffer[BUFFER_SIZE];
    size_t bytes_read;

    if (argc != 3)
    {
        printf("Usage: %s <source> <destination>\n", argv[0]);
        return 1;
    }

    source = fopen(argv[1], "rb");

    if (source == NULL)
    {
        perror("Error opening source file");
        return 1;
    }

    destination = fopen(argv[2], "wb");

    if (destination == NULL)
    {
        perror("Error opening destination file");
        fclose(source);
        return 1;
    }

    while ((bytes_read = fread(
        buffer,
        1,
        BUFFER_SIZE,
        source)) > 0)
    {
        size_t bytes_written = fwrite(
            buffer,
            1,
            bytes_read,
            destination
        );

        if (bytes_written != bytes_read)
        {
            perror("Error writing file");
            fclose(source);
            fclose(destination);
            return 1;
        }
    }

    fclose(source);
    fclose(destination);

    printf("File copied successfully\n");

    return 0;
}
```

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```
UW PICO 5.09 File: library_copy.c Modified
if (destination == NULL)
{
    perror("Error opening destination file");
    fclose(source);
    return 1;
}

while ((bytes_read = fread(
    buffer,
    1,
    BUFFER_SIZE,
    source)) > 0)
{
    size_t bytes_written = fwrite(
        buffer,
        1,
        bytes_read,
        destination
    );

    if (bytes_written != bytes_read)
    {
        perror("Error writing file");
        fclose(source);
        fclose(destination);
        return 1;
    }
}

fclose(source);
fclose(destination);

printf("File copied successfully\n");

return 0;
}
```

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```

vishalkarnati@VISHALS-MacBook-Air week9pract % nano library_copy.c
vishalkarnati@VISHALS-MacBook-Air week9pract % gcc library_copy.c -o library_copy
vishalkarnati@VISHALS-MacBook-Air week9pract % ./library_copy source.txt library_copy.txt
File copied successfully
vishalkarnati@VISHALS-MacBook-Air week9pract % cat library_copy.txt
Hello
This is a file copy test.
Operating System:
Low-level file I/O
vishalkarnati@VISHALS-MacBook-Air week9pract % time ./lowlevel_copy source.txt low.txt
File copied successfully
./lowlevel_copy source.txt low.txt  0.00s user 0.01s system 66% cpu 0.011 total

```

Task2:

Develop a program that redirects standard input and output using file descriptors and `dup2()`. Demonstrate how shells implement I/O redirection.

```

UW PIC0 5.09                                     File: redirect.c                               Modified
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int input_fd;
    int output_fd;
    pid_t pid;

    input_fd = open("input.txt", O_RDONLY);

    if (input_fd == -1)
    {
        perror("Error opening input.txt");
        return 1;
    }

    output_fd = open(
        "output.txt",
        O_WRONLY | O_CREAT | O_TRUNC,
        0644
    );

    if (output_fd == -1)
    {
        perror("Error opening output.txt");
        close(input_fd);
        return 1;
    }

    pid = fork();

    if (pid == -1)
    {
        perror("fork");
        close(input_fd);
        close(output_fd);
        return 1;
    }

    if (pid == 0)
    {
        /*
         * Redirect standard input
         */
        if (dup2(input_fd, STDIN_FILENO) == -1)
        {
            perror("dup2 stdin");
            exit(1);
        }

        /*
         * Redirect standard output
         */
        if (dup2(output_fd, STDOUT_FILENO) == -1)
        {

```

```
UW PICO 5.09 File: redirect.c Modified
}
return 1;
}

pid = fork();

if (pid == -1)
{
    perror("fork");
    close(input_fd);
    close(output_fd);
    return 1;
}

if (pid == 0)
{
    /*
     * Redirect standard input
     */
    if (dup2(input_fd, STDIN_FILENO) == -1)
    {
        perror("dup2 stdin");
        exit(1);
    }

    /*
     * Redirect standard output
     */
    if (dup2(output_fd, STDOUT_FILENO) == -1)
    {
        perror("dup2 stdout");
        exit(1);
    }

    close(input_fd);
    close(output_fd);

    /*
     * Execute cat.
     * cat reads from stdin and writes to stdout.
     */
    execlp("cat", "cat", NULL);

    perror("execlp");
    exit(1);
}

/*
 * Parent does not need these descriptors.
 */
close(input_fd);
close(output_fd);

/*
 * Wait for child to finish.
 */
waitpid(pid, NULL, 0);

printf("Redirection completed successfully\n");

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```

```
UW PICO 5.09 File: redirect.c Modified
    exit(1);
}

close(input_fd);
close(output_fd);

/*
 * Execute cat.
 * cat reads from stdin and writes to stdout.
 */
execlp("cat", "cat", NULL);

perror("execlp");
exit(1);
}

/*
 * Parent does not need these descriptors.
 */
close(input_fd);
close(output_fd);

/*
 * Wait for child to finish.
 */
waitpid(pid, NULL, 0);

printf("Redirection completed successfully\n");

return 0;
}

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```

```
vishalkarnati@VISHALS-MacBook-Air week9pract % nano redirect.c
vishalkarnati@VISHALS-MacBook-Air week9pract % nano input.txt
vishalkarnati@VISHALS-MacBook-Air week9pract % cat input.txt
Hello
This is an operating systems practical.
This file will be redirected.
Using dup2 for input and output.
vishalkarnati@VISHALS-MacBook-Air week9pract % gcc redirect.c -o redirect
vishalkarnati@VISHALS-MacBook-Air week9pract % ./redirect
Redirection completed successfully
vishalkarnati@VISHALS-MacBook-Air week9pract % cat output.txt
Hello
This is an operating systems practical.
This file will be redirected.
Using dup2 for input and output.
vishalkarnati@VISHALS-MacBook-Air week9pract %
```